Assignment-Regression Algorithmn

Problem Statement:

Answer 1: Predicting the insurance charges based on the parameters

- Input is number, so domain is learning.
- Requirements are clear, input and output fields are clear, hence it is supervised learning
- Output values are continuous numerical values, so it is regression.

Answer 2: The Basic information for this dataset is Age, Sex, BMI, Children, Smoker and the output Insurance charges. Total number of rows are 1338 rows and 6 columns (5 input and 1 output) from the given dataset.

Answer 3: Need to convert the Sex (male/female) and Smoker columns (yes/no) input fields as it is required to calculate the insurance charges. Both these data are Nominal as it is not comparable and it is like noun and we need to go for One Hot encoding.

Answer 4: Created Multiple Linear regression, Support Vector machine and Decision Tree Regressor models and deployed the finalized best model which is multiple linear regression. Below are the r2 results of each models.

1. Multiple Linear Regression
Combinations of Parameters

S.NO	fit_intercept	copy_X	tol	n_jobs	positive	r2_score
1	TRUE	TRUE	1.00E-06	None	FALSE	0.789479035
2	FALSE	TRUE	1.00E-06	None	FALSE	0.762372328
3	FALSE	FALSE	1.00E-06	None	FALSE	0.762372328
4	TRUE	TRUE	1.00E-06	1000	FALSE	0.789479035
5	TRUE	TRUE	10	1000	FALSE	0.789479035
6	TRUE	TRUE	10000	None	FALSE	0.789479035
7	TRUE	TRUE	1.00E-06	None	TRUE	0.789442939

2.Support Vector Machine regression
Combinations of Parameters

S.NO	kernel	gamma	r2_score
1	'rbf'	'scale'	-0.08843
2	poly'	'scale'	-0.0757
3	linear	'scale'	-0.0101
4	sigmoid	scale'	-0.07543
5	'rbf'	auto'	-0.08338
6	linear	auto'	-0.0101
7	poly'	auto'	-0.0757
8	sigmoid	auto'	-0.07543

3.DecisionTreeRegressor
Combination of Parameters

S.NO	criterion	splitter	max_features	r2_score
1	'squared_error'	'best'	None	0.686235
2	friedman_mse	'best'	None	0.692561
3	absolute_error	'best'	None	0.679111
4	poisson	'best'	None	0.717014
5	squared_error'	random	None	0.756403
6	friedman_mse	random	None	0.723726
7	absolute_error	random	None	0.706451
8	poisson	random	None	0.701113
9	squared_error'	random	sqrt	0.725282
10	friedman_mse	random	sqrt	0.643907
11	absolute_error	random	sqrt	0.514425
12	poisson	random	sqrt	0.613995
13	'squared_error'	'best'	log2	0.748503
14	friedman_mse	'best'	log2	0.714086
15	absolute_error	'best'	log2	0.726093
16	poisson	'best'	log2	0.750061

Answer 6:

Out of these 3 models, finalizing the model would be Multiple Linear Regression as it is maximum value nearer to 1 which is 0.789479035, other models are having less values when comparing to this model value.